

CxMP: A Linguistic Minimal-Pair Benchmark for Evaluating Constructional Understanding in Language Models

CxMP：一种用于评估语言模型构式理解能力的最小对语言基准

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Abstract

摘要

Recent work has examined language models from a linguistic perspective to better understand how they acquire language. Most existing benchmarks focus on judging grammatical acceptability, whereas the ability to interpret meanings conveyed by grammatical forms has received much less attention. We introduce the Linguistic Minimal-Pair Benchmark for Evaluating Constructional Understanding in Language Models (CxMP), a benchmark grounded in Construction Grammar that treats form-meaning pairings, or constructions, as fundamental linguistic units. CxMP evaluates whether models can interpret the semantic relations implied by constructions, using a controlled minimal-pair design across nine construction types, including the let-alone, caused-motion, and ditransitive constructions. Our results show that while syntactic competence emerges early, constructional understanding develops more gradually and remains limited even in large language models (LLMs). CxMP thus reveals persistent gaps in how language models integrate form and meaning, providing a framework for studying constructional understanding and learning trajectories in language models.

近期的研究从语言学的角度考察语言模型，以更好地理解它们如何习得语言。大多数现有的基准测试主要关注判断语法的可接受性，而对语法形式所传达意义的解读能力则关注较少。我们引入了“构式理解语言模型评估最小对基准”（Linguistic Minimal-Pair Benchmark for Evaluating Constructional Understanding in Language Models, 简称 CxMP），这是一个基于构式语法（Construction Grammar）的基准测试，将形式-意义配对（即构式）视为语言的基本单位。CxMP 通过九种构式的受控最小对设计（包括让步、致使运动和双宾语构式等），评估模型是否能够解读构式所隐含的语义关系。我们的结果显示，尽管句法能力较早出现，但构式理解的发展更为缓慢，即使在大型语言模型（LLMs）中也仍然有限。因此，CxMP 揭示了语言模型在整合形式与意义方面存在的持续性差距，为研究语言模型中的构式理解及学习轨迹提供了框架。

1 Introduction

1 引言

Recent work increasingly analyzes language models from a linguistic perspective to better understand the sources of their success and their potential for language acquisition (Hu et al., 2024; Kallini et al., 2024; Mita et al., 2025). They primarily evaluate models in terms of their grammatical acceptability, that is, whether they distinguish grammatically acceptable from unacceptable sentences based on form-level constraints (Warstadt et al., 2020, 2019; Weissweiler et al., 2025). Such methods reflect a linguistic tradition that often treats syntax as largely independent from meaning, thereby focusing on the grammaticality of linguistic expressions. However, language understanding also

近期的研究越来越多地从语言学的角度分析语言模型，以更好地理解其成功的原因及其在语言习得方面的潜力（Hu 等，2024；Kallini 等，2024；Mita 等，2025）。他们主要从语法可接受性方面来评估模型，也就是说，是否根据形式层面的限制来区分语法上可接受和不可接受的句子（Warstadt 等，2020, 2019；Weissweiler 等，2025）。这类方法反映了语言学传统中常常将句法视为与意义基本独立的倾向，从而专注于语言表达的语法正确性。然而，语言理解也

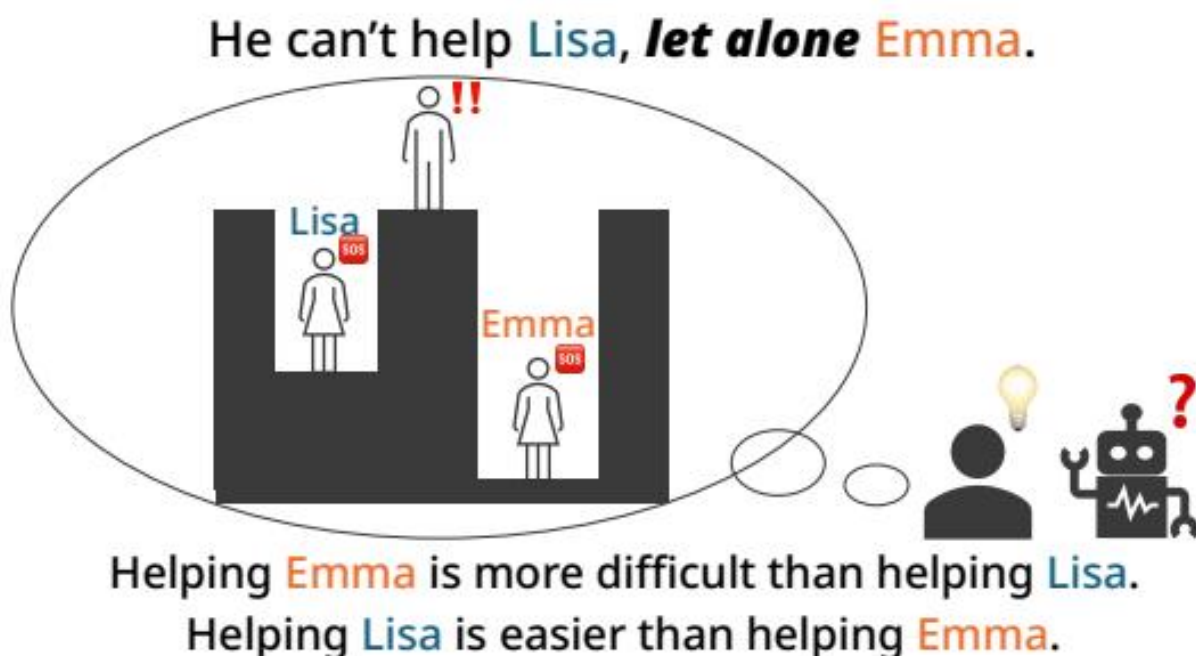


Figure 1: images/8ce5a1c85ec2fb390d276a0f6b6a12876df040041fffe9c33ec71c494568a187.jpg

Figure 1: Overview of the let-alone construction. Humans can grasp the relation between the two entities from sentences containing this construction.

图 1：让步结构概述。人类可以从包含这种结构的句子中理解这两个实体之间的关系。

requires interpreting the meanings implied by form. For instance, “Mary ran into John.” not only satisfies grammatical constraints but also conveys that “Mary” moved toward “John”, who serves as the endpoint of that motion. To bridge this gap, we adopt the framework of Construction Grammar, which treats form–meaning pairings (constructions) as the basic units of linguistic analysis. In this view, syntax and meaning are inherently interconnected, enabling analyses that focus on meanings conveyed by linguistic form.

需要解读形式所隐含的意义。例如，“Mary ran into John.” 不仅满足语法约束，还传达了“Mary” 朝向

“John” 移动，而 “John” 则是这一运动的终点。为了弥合这一差距，我们采用建构语法（Construction Grammar）的框架，该框架将形式-意义配对（建构）视为语言分析的基本单位。在这一观点下，句法和意义本质上是相互关联的，从而使分析能够专注于语言形式所传达的意义。

Previous studies have evaluated language models from a constructional perspective (Scivetti et al., 2025; Weissweiler et al., 2022; Misra and Mahowald, 2024; Potts, 2024). However, existing approaches remain limited for fine-grained linguistic analysis of constructions. They typically do not enable comprehensive and systematically controlled comparisons across models and constructions. Most focus on a single construction (Weissweiler et al., 2022; Misra and Mahowald, 2024; Potts, 2024), making it difficult to analyze relations among constructions or connect constructional understanding to other linguistic abilities. In addition, evaluation formats such as natural language inference (NLI) (Mackintosh et al., 2025; Scivetti et al., 2025) typically present a single sentence pair with a label and require a direct prediction. In such

先前的研究从构式视角评估了语言模型 (Scivetti 等, 2025; Weissweiler 等, 2022; Misra 和 Mahowald, 2024; Potts, 2024)。然而，现有方法在对构式进行细粒度语言分析时仍存在局限。它们通常无法在模型和构式之间进行全面且系统控制的比较。大多数研究集中于单一构式 (Weissweiler 等, 2022; Misra 和 Mahowald, 2024; Potts, 2024)，使得分析构式之间的关系或将构式理解与其他语言能力联系起来变得困难。此外，诸如自然语言推理 (NLI) (Mackintosh 等, 2025; Scivetti 等, 2025) 等评估形式通常提供一个带有标签的单句对，并要求直接预测。在这样的情况下，

settings, the linguistic cues driving the model’s judgments are often unclear. Moreover, these tasks are not designed to support systematic comparison across models that span from developmentally plausible models to modern large language models (LLMs). Smaller models often struggle with prompt interpretation or label classification.

设置中，驱动模型判断的语言线索往往不明确。此外，这些任务并未设计用于支持对从发展合理模型到现代大型语言模型 (LLMs) 的模型进行系统的比较。较小的模型通常在提示解释或标签分类方面存在困难。

In this work, we introduce the Linguistic Minimal-Pair Benchmark for Evaluating Constructional Understanding in Language Models (CxMP), which evaluates whether language models interpret the semantic relationships and participant roles implied by constructions. For example, consider the let-alone construction shown in Figure 1 (“He can’t help Lisa, let alone Emma.”), humans can infer that helping Emma is more difficult than helping Lisa. We go beyond existing evaluations focused on the well-formedness of sentences and instead examine whether language models grasp the meanings encoded in linguistic form. CxMP covers nine constructions, uses a controlled minimal-pair design to isolate linguistic cues, and supports evaluation across models ranging from developmentally plausible small models to large-scale LLMs.

在本工作中，我们引入了用于评估语言模型构造理解能力的句法最小对基准（Linguistic Minimal-Pair Benchmark for Evaluating Constructional Understanding in Language Models, 简称 CxMP）。该基准评估语言模型是否能够理解构造所隐含的语义关系和参与者角色。例如，考虑图 1 中所示的“let-alone”构造（“He can’t help Lisa, let alone Emma.”），人类可以推断出帮助 Emma 比帮助 Lisa 更加困难。我们超越了现有评估中仅关注句子形式正确性的局限，转而考察语言模型是否能够理解语言形式中编码的含义。CxMP 涵盖了九种构造，采用受控的最小对设计来隔离语言线索，并支持从具有发展合理性的小型模型到大规模语言模型的跨模型评估。

We evaluate a range of language models that differ in model size, data scale, and training objectives. We find that developmentally plausible small models make few correct judgments, while even the recent 70B-scale open models still struggle with some constructions, indicating substantial room for improvement in construction-level understanding. Factors influencing performance vary across construction types: formally complex constructions are more affected by model size, whereas argument-structure constructions are more sensitive to instruction tuning.

我们评估了一系列在模型规模、数据量和训练目标方面不同的语言模型。我们发现，发展上合理的较小模型做出的正确判断较少，而即使是最近的 700 亿参数级别的开放模型在某些构造上仍存在困难，这表明在构造层面的理解上仍有很大的提升空间。影响性能的因素在不同构造类型之间有所不同：形式上复杂的构造更容易受到模型规模的影响，而论点结构构造则更敏感于指令调优。

To demonstrate the usefulness of our dataset for comparison with human language acquisition, we further analyze learning trajectories and pseudoword generalization. Unlike BLiMP, which evaluates sentence acceptability, CxMP tests whether models interpret meanings encoded in syntactic form. Models reach high BLiMP accuracy early in training, whereas CxMP performance improves gradually. A fine-grained analysis of CxMP reveals stage-like learning transitions from near-random responses to heuristic use and eventual stabilization. Under a more controlled condition where lexical items are replaced with pseudowords, models, unlike humans, show a substantial decline in constructional meaning interpretation.

为了展示我们的数据集在与人类语言习得进行比较时的实用性，我们进一步分析了学习轨迹和伪词泛化能力。与评估句子可接受性的 BLiMP 不同，CxMP 测试模型是否能够理解语法形式中编码的含义。模型在训练初期就能达到较高的 BLiMP 准确率，而 CxMP 的表现则逐渐提升。对 CxMP 的细致分析揭示了从接近随机的回应到启发式使用，最终达到稳定的学习阶段的转变过程。在一种更受控的条件下，即用伪词替换词汇项时，模型在构造意义理解方面表现出显著的下降，而人类则不会出现这种情况。

Taken together, these findings highlight the need for evaluation beyond traditional grammatical acceptability to capture the semantic contributions of constructions. They further demonstrate that the proposed dataset provides a foundation for future research on constructional understanding.

综合这些发现，这些结果突显了超越传统语法可接受性进行评估的必要性，以捕捉构式的意义贡献。它们进一步表明，所提出的语料库为未来关于构式理解的研究提供了基础。

2 Motivations

2 动机

In this section, we outline four key properties that an evaluation dataset should satisfy to assess constructional understanding in language models effectively.

在本节中，我们概述了四个关键属性，这些属性应被评估数据集满足，以便有效地评估语言模型的构式理解能力。

Model Agnostic Recent work has examined developmentally plausible small models to investigate language acquisition under limited data conditions (Hu et al., 2024). Evidence from large language models, however, shows that generalization can arise even without strong constraints, driven by inherent simplicity biases that offer insights for linguistic theory (Futrell and Mahowald, 2025). Most existing studies rely on NLI (Mackintosh et al., 2025; Scivetti et al., 2025), where success presupposes understanding task-specific notions such as entailment and contradiction, making zero-shot evaluation difficult for smaller models that struggle with prompt interpretation and label classification. Fine-tuning further obscures the boundary between pretrained and newly learned knowledge. Hence, there is a need for benchmarks that enable consistent evaluation across both small and large language models.

模型无关的近期研究探讨了在有限数据条件下开发出具有发展合理性的小型模型，以研究语言习得 (Hu 等, 2024)。然而，来自大型语言模型的证据表明，即使没有强约束，泛化也可以出现，这是由内在的简单性偏见驱动的，这些偏见为语言理论提供了见解 (Futrell 和 Mahowald, 2025)。大多数现有研究依赖于自然语言推理 (NLI) (Mackintosh 等, 2025; Scivetti 等, 2025)，其中成功预设了对诸如蕴含和矛盾等任务特定概念的理解，这使得对于那些在提示解释和标签分类方面存在困难的小型模型而言，零样本评估变得困难。微调进一步模糊了预训练知识和新学知识之间的界限。因此，需要一种基准，以实现对小型和大型语言模型的一致评估。

Comprehensive across Constructions Most prior studies focus on individual constructions such as the Comparative Correlative (Weissweiler et al., 2022), AANN (Misra and Mahowald, 2024), or Preposing in PiPP (Potts, 2024), examining how language models recognize, reason about, and generalize over these specific patterns. However, Construction Grammar encompasses a wide range of constructions, from lexically grounded substantive ones to highly abstract schematic

ones. A benchmark that systematically covers diverse constructions enables more comprehensive analyses of the relationships among constructions and their connections to other aspects of linguistic knowledge.

跨构式的全面性

大多数先前的研究都聚焦于个别构式，例如比较相关构式（Weissweiler 等，2022），AANN（Misra 和 Mahowald，2024），或 PiPP 中的前置（Potts，2024），探讨语言模型如何识别、推理以及在这些特定模式上进行泛化。然而，构式语法涵盖了从以词汇为基础的实体构式到高度抽象的图式构式的广泛范围。一个系统性覆盖多样构式的基准，能够更全面地分析构式之间的关系及其与其他语言知识方面的联系。

Controlled Design Datasets such as NLI and CoLA, which consist of sentences paired with labels, allow for diverse forms of evaluation but risk inducing shallow heuristics or surface-level biases in model judgments. Moreover, the rationale for labeling is often implicit, making it unclear which

受控设计的数据集（如 NLI 和 CoLA），这些数据集由句子与标签配对组成，允许进行多样化的评估，但可能会导致模型判断中出现浅层启发式方法或表面偏见。此外，标注的依据往往具有隐含性，使得不清楚哪些

linguistic cues the model relies on to make its judgments. For example, sentences with high vocabulary overlap are more likely to be classified as entailment, whereas those containing negation tend to receive negative labels. To mitigate such issues, a minimal-pair design that isolates controlled differences is desirable, enabling precise evaluation of which linguistic factors models are actually sensitive to. Furthermore, common nouns, while natural and frequent, often exhibit strong lexical associations; for instance, teacher co-occurs with teach far more often than with learn. To control for such lexical biases, it is important to introduce multiple noun variants and systematically alternate the roles or relationships of participants (e.g., swapping subject and object), allowing a more fine-grained analysis of models' underlying biases.

模型在做出判断时所依赖的语言线索。例如，词汇重叠程度高的句子更有可能被归类为蕴含关系，而包含否定的句子则往往获得负面标签。为了解决此类问题，采用最小对设计（minimal-pair design）以隔离可控的差异是理想的，这使得能够精确评估模型实际对哪些语言因素敏感。此外，普通名词虽然自然且常见，但往往表现出强烈的词汇关联；例如，“teacher”（教师）与“teach”（教学）共同出现的频率远高于与“learn”（学习）共同出现的频率。为了控制此类词汇偏见，引入多种名词变体并系统地交替参与者角色或关系（例如，交换主语和宾语）是非常重要的，这有助于更细致地分析模型内在的偏见。

Multiple Sentence Single-sentence minimal pairs are effective for evaluating the formal naturalness of sentences but are insufficient for assessing the semantic contributions of constructions. A multi-sentence design is therefore necessary. Among existing datasets adopting such a design, EWoK serves as a representative example: it evaluates world knowledge through pairs consisting of a context describing a situation and a target proposition whose plausibility is judged relative to that context. However, benchmarks that apply this framework to the study of language acquisition in language models from a linguistic perspective remain limited. Since understanding a sentence's meaning may involve inferences drawn from its surrounding context during training, adopting a multisentence minimal-pair design enables investigation of how well language models capture semantic relations between sentences

多句的单句最小对(minimal pairs)对于评估句子的正式自然度是有效的,但对于评估构式(constructions)的语义贡献则显得不足。因此,有必要采用多句设计。在现有的采用此类设计的数据集中,EWoK 是一个具有代表性的例子:它通过由描述情境的上下文和一个目标命题组成的配对来评估世界知识,判断目标命题相对于上下文的合理性。然而,从语言学视角出发,将这一框架应用于语言模型中语言习得研究的基准仍然有限。由于在训练过程中理解句子的含义可能涉及从其周围上下文中进行推理,采用多句最小对设计能够研究语言模型在多句之间捕捉语义关系的能力。

In this work, we construct CxMP, a dataset that integrates all of these motivations into a unified framework.

在本工作中，我们构建了 CxMP 数据集，将所有这些动机整合到一个统一的框架中。

3 CxMP

3 CxMP

We assume that when a model correctly identifies the meaning encoded in a construction and the roles or relations of its entities, it demonstrates constructional understanding through successful form–meaning mapping. CxMP consists of sentence pairs (Evaluation Pairs), each comprising a sentence containing a target construction (Constructional Example) and a sentence probing the meaning mapped to that construction (Diagnostic Sentence). Examples for each construction are shown in Table 1, where N1 and N2 denote the entities involved. Each Diagnostic Sentence has two minimally different versions, plausible and implausible. Each construction also has two variants (A/B) that probe the construction’s implied meaning and relational roles from different perspectives. To maintain temporal consistency, all sentences are written in the past tense.

我们假设，当模型能够正确识别一个结构中所编码的意义以及其中实体的角色或关系时，它便通过成功的形式–意义映射展示了结构理解能力。CxMP 包含句子对（评估对），每一对包括一个包含目标结构（结构性例子）的句子和一个探查映射到该结构的意义的句子（诊断句子）。每个结构的例子如表 1 所示，其中 N1 和 N2 表示涉及的实体。每个诊断句子有两个最小不同的版本，合理和不合理。每个结构还有两个变体（A/B），从不同角度探查该结构所隐含的意义和关系角色。为了保持时间一致性，所有句子均使用过去时态书写。

3.1 Constructions

3.1 构造

In this work, following Bonial and Tayyar Madabushi (2024), we adopt nine types of constructions. Examples of these constructions are shown in Table 1. The constructions range from partially fixed patterns to fully schematic ones.

在本研究中，我们遵循 Bonial 和 Tayyar Madabushi (2024) 的做法，采用九种类型的结构。这些结构的示例如表 1 所示。这些结构从部分固定模式到完全图式化结构不等。

LET-ALONE In the fixed let alone construction, the two conjuncts represent different points on the same scalar dimension, where the second clause denotes a stronger degree • typically a stronger form of negation, than the first. This construction has been analyzed in detail by Fillmore (Fillmore et al., 1988) and others, as it exhibits unique properties combining a negative polarity environment and a comparative relation. In this dataset, we assume that actions at a higher point on the negative scale are harder to perform and design sentences that ask which action is more difficult. While a wide range of syntactic elements (nouns, verbs, clauses) can appear around let alone, we focus on the simplest and most controllable case: nominal comparison. Because the construction requires a licenser for negative polarity, we explicitly include negation in the stimuli using negative auxiliaries such as “cannot” and “don’t”. Although Bonial and Tayyar Madabushi (2024) includes the MUCH-LESS, it differs from let alone only in surface form and is therefore omitted here.

独让结构 (let-alone construction) 中，两个并列成分表示同一标量维度上的不同点，其中第二个从句表示的程度通常比第一个更强，通常是更强形式的否定。Fillmore (Fillmore et al., 1988) 及其他学者对此结构进行了详细分析，因为该结构结合了否定极性环境和比较关系，具有独特的性质。在本数据集中，我们假设在否定标尺上位置更高的动作更难完成，并设计句子询问哪个动作更困难。虽然围绕 “let alone” 可以出现各种句法成分（名词、动词、从句），但我们关注最简单且最可控的情况：名词比较。由于该结构需要否定极性许可者，我们在刺激材料中明确使用否定助动词（如 “cannot” 和 “don’t”）来包含否定。虽然 Bonial 和 Tayyar Madabushi (2024) 包含了 MUCH-LESS，但其与 “let alone” 仅在表层形式上不同，因此在此省略。

CAUSATIVE-WITH This construction is characterized by a with-phrase that follows the direct object. The direct object denotes the affected entity, and the with-phrase indicates what is applied

or provided to it. In our dataset, we restrict both the subject and the object to human nouns, allowing us to test whether the model can correctly interpret the supply–recipient relation between entities.

CAUSATIVE-WITH 这种结构的特点是直接宾语之后跟着一个 with-短语。直接宾语表示受影响的实体，而 with-短语则表明施加或提供给它的内容。在我们的数据集中，我们将主语和宾语都限制为人类名词，这样我们就可以测试模型是否能够正确理解实体之间的供应–接受关系。

WAY-MANNER This construction expresses movement achieved by creating or carving out one's own path, often despite external difficulty or through indirect means (Goldberg, 1995). Formally, a fixed phrase one's way appears in the

WAY-MANNER 这一构式表达了通过创造或开辟自己的道路来实现的运动，通常是在外部困难的情况下或通过间接手段实现的（Goldberg, 1995）。在形式上，一个固定短语 one's way 出现在文本中

Construction	Constructional Example	Diagnostic Sentence
LET-ALONE	He can't help N1, let alone N2.	Helping N2/N1 is more difficult than helping N1/N2.Helping N1/N2 is easier than helping N2/N1.
CAUSATIVE-WITH	N1 covered N2 with a coat.	The one who provided the coat is N1/N2.The one who received the coat is N2/N1.
WAY-MANNER	N1 pushed his way into the room.	He did so while moving/staying.He did so with/without effort.
COMPARATIVE-CORRELATIVE (CC)	The harder you work,the stronger you become.N1 works harder than N2.	The one who becomes stronger is N1/*N2.
The stronger you become,the harder you work.N1 becomes stronger than N2.	The one who works harder is N1/*N2.	
CONATIVE	N1 kicked at the ball.	Whether she succeeded in kicking the ball is unclear/clear.Whether the ball was affected is unclear/clear.
DITRANSITIVE	N1 bought N2 a gift.	The one who gave the gift is N1/N2.The one who received the gift is N2/N1.
CAUSED-MOTION	N1 pulled N2 out of the water.	The one who moved her is N1/N2.The one who got moved is N2/N1.
RESULTATIVE	N1 pushed N2 awake.	The one who caused her to become awake is N1/N2.The one who became awake is N2/N1.
INTRANSITIVE-MOTION	N1 ran into N2.	The one who moved is N1/N2.The one who he moved toward is N2/N1.

LET-ALONE	他无法帮助 N1，更不用说 N2 了。	帮助 N2/N1 比帮助 N1/N2 更
致使	N1 用外套盖住了 N2。	提供外套的人是 N1/N2。接
方式-方式	N1 推开人群进入房间。	他一边移动/保持原状一边这
比较-相关 (CC)	你越努力，就会变得越强大。N1 比 N2 更努力。	变得更强的是 N1/*N2。
你越强，就越要努力。N1 比 N2 更强大。	工作更努力的是 N1/*N2。	
动词性（表示动作或意图的）	N1 踢了球。	她是否成功踢到了球还不清楚
不及物动词	N1 给 N2 买了一份礼物。	给予礼物的人是 N1/N2。接
CAUSED-MOTION	N1 将 N2 拉出了水面。	使她移动的是 N1/N2。被移
结果性	N1 推醒了 N2。	使她觉醒的人是 N1/N2。觉
不及物动词	N1 与 N2 相撞。	移动的是 N1/N2。他移动的

Table 1: Constructions that we focus on. * indicates implausible versions.

表 1: 我们关注的构式。* 表示不可行的版本。

direct-object position, followed by a prepositional phrase that indicates the path. To evaluate the understanding of this construction, we design sentences asking whether the event involves motion or rest and whether it is carried out with strong intentionality.

直接宾语位置，随后是表示路径的介词短语。为了评估对这种结构的理解，我们设计句子，询问事件是否涉及运动或静止，以及是否具有强烈的意图性。

COMPARATIVE-CORRELATIVE This construction has been extensively studied as one of the most challenging phenomena to explain within the generative framework and as one of the most prominent constructions in English (Fillmore, 1986; Hoffmann, 2017). Recent probing studies have also examined how language models understand this construction (Weissweiler et al., 2022). It consists of two clauses, both beginning with the + comparative forms of adjectives or adverbs (Goldberg, 2003). The second clause functions as a dependent variable determined by the first, often implying causal or temporal relationships between the two. Following Weissweiler et al. (2022), our dataset uses Constructional Examples in which the first clause establishes a relational condition between entities, and the subsequent sentence asks which entity exhibits the resulting dependent property.

比较-相关结构这一结构已被广泛研究，被认为是生成框架内最难解释的现象之一，同时也是英语中最显著的结构之一（Fillmore, 1986; Hoffmann, 2017）。最近的探索性研究还考察了语言模型如何理解这一结构（Weissweiler 等, 2022）。该结构由两个分句组成，两个分句均以形容词或副词的比较级形式开头（Goldberg, 2003）。第二个分句作为由第一个分句决定的因变量，通常暗示着两个分句之间存在因果或时间关系。根据 Weissweiler 等（2022）的研究，我们的数据集使用了构造性例子，其中第一个分句在实体之间建立了一种关系条件，随后的句子则询问哪一个实体表现出由此产生的从属性。

CONATIVE The following five constructions are those identified by Goldberg (1995) as representative argument-structure constructions. The conative construction features a direct object preceded by at, indicating that the agent attempts or makes

致使类（Conative）以下五种结构是 Goldberg（1995）所识别的具有代表性的论元结构构造。致使类结构的特点是有一个由介词 at 引导的直接宾语，表明施事者试图或做出某种动作。

an effort to act upon the target, without implying success. In our dataset, half of the Constructional Examples instantiate the conative construction and the other half use standard transitive sentences. In the Diagnostic Sentences, we evaluate on whether models can distinguish that in conative cases the result of the action is uncertain, whereas in transitive cases the action is successful and the object is affected.

一种努力去实施行动，而不暗示成功。在我们的数据集中，一半的构式例子实例化了动词构式，而另一半则使用标准的及物句。在诊断句子中，我们评估模型是否能够区分，在动词构式的情况下，行动的结果是不确定的，而在及物句的情况下，行动是成功的，并且对象受到了影响。

DITRANSITIVE This construction follows the schema SUBJ V OBJ1 OBJ2, expressing that the subject transfers or gives OBJ2 (the theme) to OBJ1 (the recipient). We evaluate whether models capture this transfer-of-possession meaning by asking who gave and who received the item in the sentence.

DITRANSITIVE 这种结构遵循“主语 + 谓语 + 宾语 1 + 宾语 2”的模式，表示主语将宾语 2（主题）转移或给予宾语 1（接受者）。我们通过询问句子中是谁给予了物品、是谁接收了物品来评估模型是否能够捕捉到这种所有权转移的含义。

CAUSED-MOTION This construction expresses that the subject (causer) causes the theme to move along a path indicated by an oblique phrase. Formally, it follows the schema SUBJ V OBJ OBL, where the oblique phrase always denotes direction or path. We evaluate whether models correctly assign causal semantic roles by asking Diagnostic Sentences, such as who caused the motion and who was moved.

CAUSED-MOTION 这种构式表示主语（施事者）使事主沿斜格短语所指示的路径移动。形式上，它遵循 SUBJ V OBJ OBL 的结构，其中斜格短语始终表示方向或路径。我们通过提问诊断性句子（如谁导致了运动，谁被移动了）来评估模型是否正确地分配了因果语义角色。

RESULTATIVE This construction follows the pattern SUBJ V OBJ XCOMP, indicating that the subject's (agent's) action causes the object (patient) to

结果性这一结构遵循主语（施事者）+ 动词 + 宾语 + 宾补的模式，表明主语（施事者）的行为导致宾语（受事者）

undergo a change of state (XCOMP). The XCOMP is typically an adjective phrase (e.g., flat, open) or prepositional phrase denoting the resulting state. We evaluate whether models can identify which participant underwent the change and which one caused it, thereby testing their understanding of the causal and resultative semantics inherent to the construction.

经历状态变化 (XCOMP)。通常情况下，XCOMP 是一个形容词短语（例如：flat, open）或介词短语，表示结果状态。我们评估模型是否能够识别出哪一参与者经历了状态变化，以及哪一参与者导致了这一变化，从而测试它们对这一结构所固有的因果关系和结果关系语义的理解。

INTRANSITIVE-MOTION This construction follows the schema SUBJ V OBL and denotes that the subject (agent) moves autonomously. In our dataset, the oblique phrase also denotes an animate entity, and Diagnostic Sentences ask which participant moved and which one the mover arrived at, assessing whether the model understands directionality and goal relations.

不及物动词短语 This construction follows the schema SUBJ V OBL and denotes that the subject (agent) moves autonomously. In our dataset, the oblique phrase also denotes an animate entity, and Diagnostic Sentences ask which participant moved and which one the mover arrived at, assessing whether the model understands directionality and goal relations.

3.2 Data Generation with LLMs

3.2 使用大语言模型生成数据

We automatically generate Constructional Examples for each construction using gpt-5-chat-latest based on predefined templates designed to elicit a corresponding Diagnostic Sentence.¹ For each construction, we prompt the model to fill slot variables (e.g., subject, oblique) with contextually appropriate words, and use multiple random seeds to enhance lexical diversity. We then filter the generated sentences and pair them with corresponding plausible and implausible diagnostic

statements, ensuring that the final dataset includes only semantically coherent form–meaning mappings. We manually inspect the resulting dataset to ensure its quality and validity. In total, we finally create 43k Evaluation Pairs.

我们基于预定义的模板，使用 gpt-5-chat-latest 自动生成每个构式的构造性例子，这些模板旨在引发相应的诊断句子。¹ 对于每个构式，我们提示模型用语境合适的词语填充槽位变量（例如主语、宾语），并使用多个随机种子来增强词汇多样性。然后，我们过滤生成的句子，并将其与相应的合理和不合理诊断陈述配对，确保最终数据集仅包含语义上连贯的形式-意义映射。我们手动检查最终的数据集以确保其质量和有效性。总计，我们最终创建了 43k 个评估对。

Human Data Validation We conduct a human validation study using Prolific with native English speakers. Participants read a preceding sentence and select the continuation that is more plausible. We collect three independent annotations per item. Across 896 items, the majority-vote accuracy reaches 96.65%, and full agreement among annotators reaches 83.59%, suggesting that the automatically generated items are generally reliable. The number of items varies by construction (typically 112, and 56 for constructions without switched variants). Detailed per-construction results and annotation statistics are provided in Appendix D.

人工数据验证 我们使用 Prolific 进行人工验证研究，参与者为母语为英语的用户。参与者阅读一个前导句子，并选择更合理的延续内容。我们对每个项目收集三个独立的标注。在 896 个项目中，多数投票准确率达到了 96.65%，标注者之间完全一致的比例达到了 83.59%，这表明自动生成的项目总体上是可靠的。项目的数量因构造方式而异（通常为 112 个，对于没有切换变体的构造方式则为 56 个）。附录 D 中提供了详细的每种构造结果和标注统计信息。

Entities and Generation Procedure We used four types of names that represent entities, Fe-

实体和生成过程我们使用了四种代表实体的名称，Fe-

male name, Male name, and Name+alphabet (e.g., Name B), and common noun.² While common nouns represent the most natural expressions, proper-name variants mitigate lexical biases, and name–alphabet forms provide the most controlled condition. For constructions involving two entities, we also created switched versions by reversing their positions to test whether models rely on shallow heuristics. They correspond to the sentence swapping N1 and N2 in Table 1.

男性名字，男性名字，以及姓名 + 字母（例如，姓名 B），以及普通名词。² 虽然普通名词代表了最自然的表达方式，但专有名词变体可以减轻词汇偏见，而姓名–字母形式提供了最受控制的条件。对于涉及两个实体的结构，我们还通过反转它们的位置来创建了交换版本，以测试模型是否依赖于浅层启发式规则。它们对应于表 1 中句子交换 N1 和 N2 的情况。

4 Experiment

4 实验

4.1 Settings

4.1 设置

Models We use developmentally appropriate models from the BabyLM Challenge (10M/100M tokens) (Charpentier et al., 2025), trained on child-directed corpora such as CHILDES, and three BabyBERTa models trained with different random seeds (Huebner et al., 2021). To examine data size effects, we use Open-sci-ref-0.01 (Nezhurina et al., 2025), which provides 1.3B and 1.7B models trained on C4 (50B/300B tokens) and Nemotron (300B/1T tokens), respectively. For model size effects, we use the Pythia suite (Biderman et al., 2023), which includes ten models from 14M to 12B parameters. To compare training objectives, we contrast causal and masked language modeling using GPT-2-medium, RoBERTa-base/large, and Pythia-410m. Their sizes are comparable, while data sizes increase in the order GPT-2 (40G) < RoBERTa (160G) < Pythia (825G). For larger

open LLMs, we use Llama-3.1 (Meta, 2024) (8B/70B, base and instruction-tuned) to assess the effect of instruction tuning. Finally, we include GPT-5³, a widely used closed-source commercial model, as an approximate upper bound for open models.

我们使用了来自 BabyLM 挑战赛 (10M/100M tokens) (Charpentier 等人, 2025) 的发展适配模型, 这些模型是在儿童导向语料库 (如 CHILDES) 上进行训练的, 并且使用了三个不同随机种子训练的 BabyBERTa 模型 (Huebner 等人, 2021)。为了研究数据量的影响, 我们使用了 Open-sci-ref-0.01 (Nezhurina 等人, 2025), 该模型提供了分别在 C4 (50B/300B tokens) 和 Nemotron (300B/1T tokens) 上训练的 1.3B 和 1.7B 模型。为了研究模型规模的影响, 我们使用了 Pythia 套件 (Biderman 等人, 2023), 其中包括从 14M 到 12B 参数的十个模型。为了比较训练目标, 我们通过 GPT-2-medium、RoBERTa-base/large 和 Pythia-410m 对因果语言建模和掩码语言建模进行了对比。它们的规模相当, 而数据量按 GPT-2 (40G) < RoBERTa (160G) < Pythia (825G) 的顺序递增。对于更大的开放型大语言模型, 我们使用了 Llama-3.1 (Meta, 2024) (8B/70B, 基础版和指令调优版) 来评估指令调优的效果。最后, 我们包含了 GPT-5³, 这是一种广泛使用的闭源商业模型, 作为开放模型的近似上限。

Evaluation To evaluate whether each language model grasps the construction and its meaning, we computed the probabilities of the Plausible Evaluation Pair and Implausible Evaluation Pair for each set. We use length-normalized log probability for CLM and pseudo log-likelihood (Salazar et al., 2020) for MLM. In contrast, GPT-5 is not directly accessible to probabilities; we estimate its likelihoods through prompting. Details of the evaluation procedure are in Appendix E.

评估为了评估每个语言模型是否理解了构造及其含义, 我们计算了每个数据集的合理评估对和不合理评估对的概率。我们使用长度归一化的对数概率 (log probability) 来评估 CLM 模型, 并使用伪对数似然 (Salazar 等, 2020) 来评估 MLM 模型。相比之下, GPT-5 无法直接获取概率; 我们通过提示 (prompting) 来估计其似然值。评估过程的详细信息见附录 E。

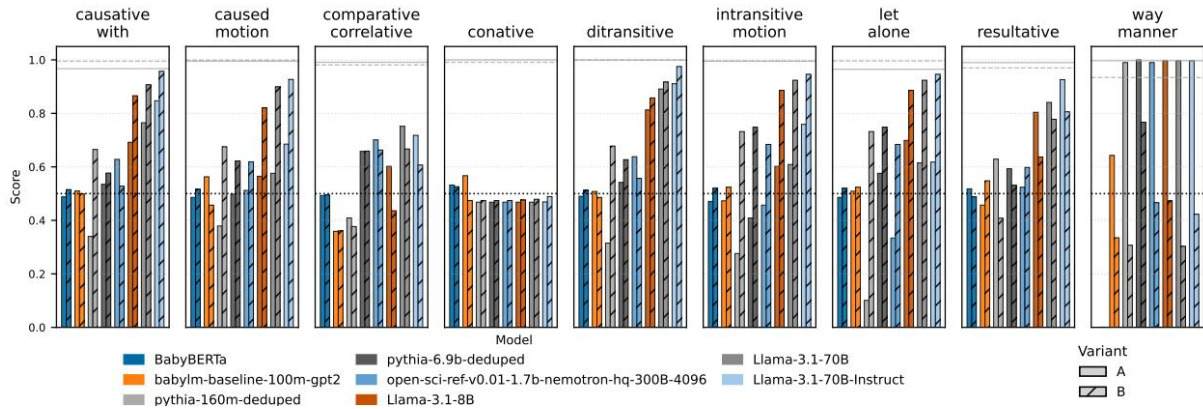


Figure 2: images/a45c9856240314708c20727e7ce98674fc05545ad3b174fa4db9e95bff30aeb0.jpg

Figure 2: Scores of each language model across constructions and variants. Solid and dashed lines at the top of the figure represent the scores of GPT-5 for variants A and B, respectively.

图 2: 各语言模型在不同构造和变体上的得分。图顶部的实线和虚线分别表示 GPT-5 在变体 A 和 B 上的得分。

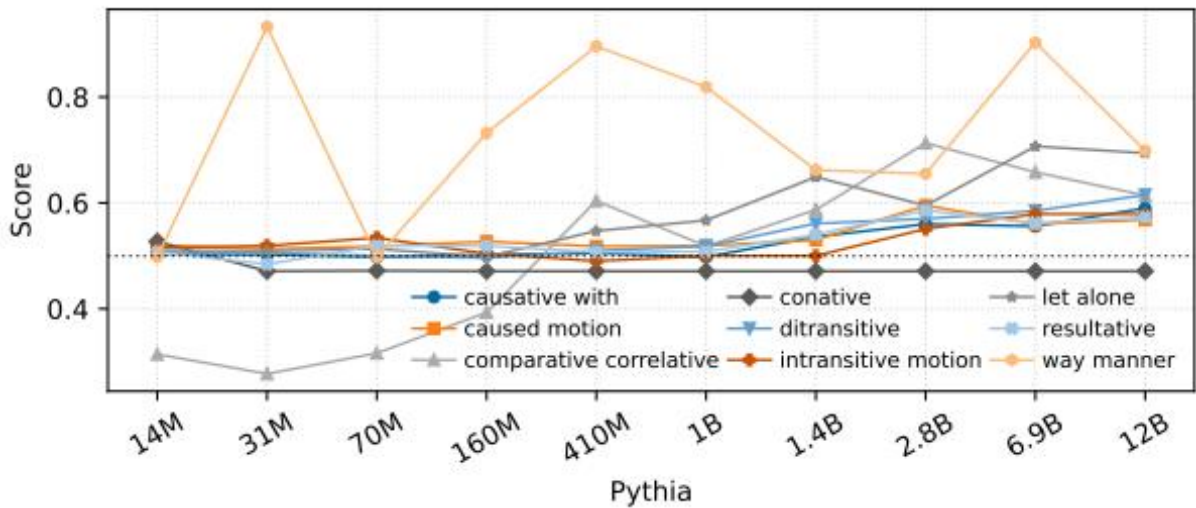


Figure 3: images/a64c5edeeb051dc14228e9bec474f0b66b2ddf8549b648ac4d35d7d0cd7879df.jpg

Figure 3: Scores of Pythia models by model size.

图 3: Pythia 模型的得分与模型大小。

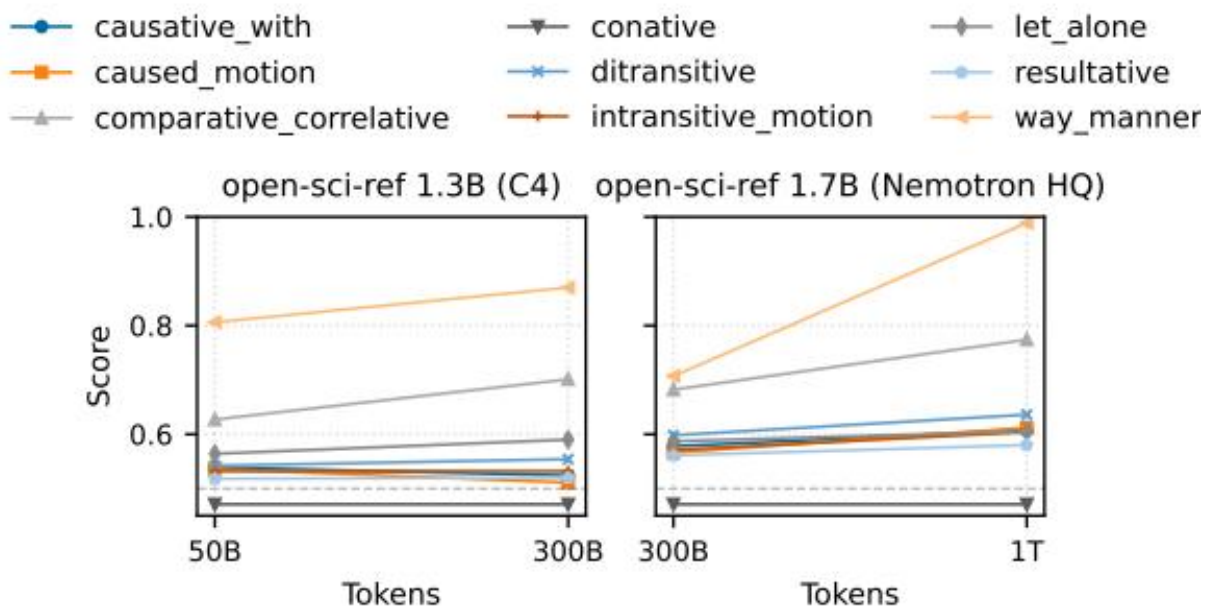


Figure 4: images/c502267fb1a1e9705e8427f6afbf5a9f6c025a7c4b848691d667013a20038678.jpg

Figure 4: Scores of open-sci-ref models by data size.

图 4: 按数据量大小划分的开放科学参考模型得分。

4.2 Results

4.2 结果

Overall We evaluate eight models of varying sizes from the series used in this study, as shown in Figure 2. Small-scale models trained on developmentally appropriate corpora perform at a near-chance level. While previous work has shown that even small models can perform reasonably well on formal grammatical benchmarks such as BLiMP, our results indicate that understanding constructional meaning is considerably more challenging. In contrast, the large closed-source GPT-5 accurately interprets most constructions. However, even open models with 70B parameters continue to struggle with certain constructions, and some remain entirely unsolved. Mid-sized models, in particular, show a mix of very high and very low accuracies across constructions, suggesting reliance on specific heuristics or biases. A detailed analysis of these biases is provided in Section 5.2.

总体而言，我们评估了本研究中使用的系列模型中不同规模的八种模型，如图 2 所示。在发育适宜的语料库上训练的小型模型的表现接近随机水平。虽然之前的研究表明，即使是小型模型也能在诸如 BLiMP 等正式语法基准测试中表现得相当不错，但我们的结果表明，理解构式意义要困难得多。相比之下，大型闭源模型 GPT-5 能够准确解释大多数构式。然而，即使是有 700 亿参数的开放模型，仍然在某些构式上存在困难，有些甚至完全未被解决。特别是中型模型，在构式上的表现呈现出极高和极低准确率的混合，这表明它们依赖于特定的启发式方法或偏见。这些偏见的详细分析在第 5.2 节中提供。

Data/Model Size Figure 3 shows the accuracy of each construction across ten model sizes in the Pythia series. Overall, except for the CONATIVE, most constructions exhibit a clear upward trend in scores as model size increases. In particular, non-standard constructions such as LET-ALONE and

数据/模型规模图 3 展示了 Pythia 系列中十种模型规模下每种构造的准确性。总体而言，除了 CONATIVE 之外，大多数构造在模型规模增加时表现出明显的分数上升趋势。特别是像 LET-ALONE 这样的非常规构造，

CC show a strong dependence on model expressivity, with substantial performance improvements observed only in larger models. For the remaining constructions, scores begin to rise around the 1.4B parameter scale, while smaller models remain near chance level. In contrast, WAY-MANNER shows large fluctuations across model sizes, with no consistent trend.

CC 显示出对模型表达能力的强烈依赖，只有在更大的模型中才观察到显著的性能提升。对于其余的构造，分数在约 1.4B 参数规模时开始上升，而较小的模型则保持在随机水平附近。相比之下，WAY-MANNER 在不同模型规模下表现出较大的波动，没有一致的趋势。

Figure 4 presents results from the Open-sci-ref-0.01 models trained on different data scales. On average, performance improves with larger training datasets, though the gain from 50B to 300B tokens remains limited. At the level of individual constructions, most show minimal change, or even slight decreases, between 50B and 300B. However, all constructions exhibit score increases between 300B and 1T, suggesting that, under this setup, larger-scale training data substantially contribute to improvements in constructional meaning understanding.

图 4 展示了在不同数据规模上训练的 Open-sci-ref-0.01 模型的结果。总体而言，随着训练数据集的增大，性能有所提升，但从 500 亿到 3000 亿个 token 的提升效果有限。在个体结构层面，大多数结构在 500 亿到 3000 亿个 token 之间变化不大，甚至略有下降。然而，所有结构在 3000 亿到 1 万亿个 token 之间都表现出分数的提升，这表明，在这种设置下，更大规模的训练数据显著促进了对结构意义的理解。

Instruction Tuning Figure 5 shows the scores of the base and instruction-tuned variants of Llama3.18B/70B. Overall, the instruction-tuned models tend to achieve higher scores than their base counter-

图 5 展示了 Llama3.18B/70B 的基础模型和指令调优版本的得分。总体而言，指令调优模型的得分通常高于其基础版本。

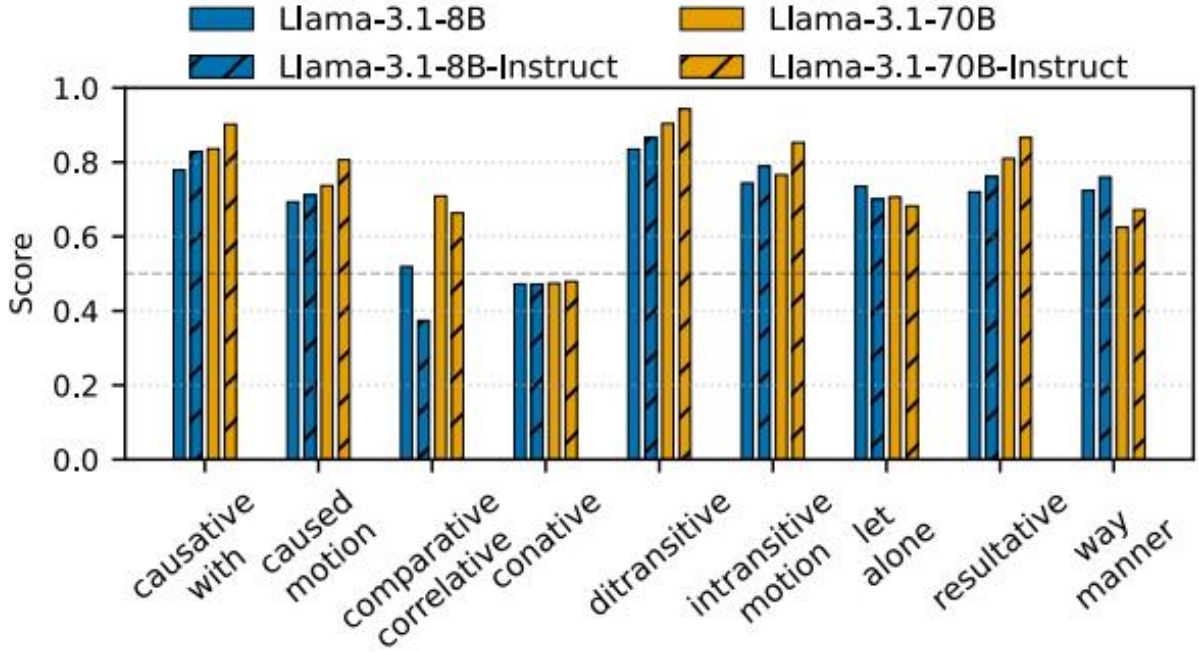


Figure 5: images/a06ee7820ab868080d179fdb447eede8807cacc836e40bc9b9b0bece9497ba1.jpg

Figure 5: Scores of Llama3.1 base/instruction models.

图 5: Llama3.1 基础/指令模型得分。

Model	Model sz.	Data sz.	Obj.	Score
RoBERTa-base	125M	160G	MLM	0.573
GPT-2-medium	345M	40G	CLM	0.537
RoBERTa-large	355M	160G	MLM	0.674
Pythia-410m	410M	825G	CLM	0.544

模型	模型号	数据大小。	目标	得分
RoBERTa-base	125M	160G	MLM	0.573
GPT-2 中型	345M	40G	CLM	0.537
RoBERTa-large	355M	160G	MLM	0.674
Pythia-410m	410M	825G	CLM	0.544

Table 2: Scores of CLM & MLM models. Obj. indicates objective and Sz. indicates size.

表 2: CLM 与 MLM 模型的得分。Obj. 表示目标, Sz. 表示规模。

parts. However, the magnitude of this effect varies across constructions: while consistent improvements are observed for argument-structure constructions, non-standard constructions such as LETALONE and CC show a decrease in performance. These results suggest that instruction tuning does not necessarily enhance all aspects of constructional meaning understanding.

然而，这种效应的强度在不同的构式中有所不同：虽然对于论元结构构式观察到持续的改进，但对于非标准构式（如 LETALONE 和 CC）则表现出性能下降。这些结果表明，指令调优并不一定增强所有构式意义理解方面的表现。

Objectives Table 2 compares RoBERTa and CLM models of comparable size. Although RoBERTa-large is slightly larger, it outperforms Pythia despite Pythia being trained on over five times more data. Similarly, RoBERTa-base achieves higher scores than the Pythia model, exceeding it in both model and data size. Overall, within the examined range, MLMs tend to outperform CLMs. Recent theoretical work (Zhang et al., 2024) suggests that MLMs induce richer co-occurrence patterns and stronger semantic associations, enhancing classification performance, whereas their fixed masking rate limits generalization in generative tasks with variable-length inputs. Although our task differs from probability-based classification, the results indicate that MLMs may exploit bidirectional context to more effectively integrate form and meaning. Consistent with this trend, several BabyLM Challenge systems (Hu et al., 2024) have adopted hybrid MLM-CLM objectives (Yu et al., 2024; Charpentier and Samuel, 2024). A systematic exploration of which linguistic abilities particularly benefit from the MLM objective remains a promising direction for future work.

表 2 比较了大小相近的 RoBERTa 和 CLM 模型。虽然 RoBERTa-large 稍大一些，但它在 Pythia 上的表现更好，尽管 Pythia 是在超过五倍的数据上进行训练的。同样地，RoBERTa-base 的得分也高于 Pythia 模型，在模型规模和数据规模上都超过了 Pythia。总体而言，在所研究的范围内，MLMs 通常优于 CLMs。最近的理论研究（Zhang 等，2024）表明，MLMs 能够诱导出更丰富的共现模式和更强的语义关联，从而提升分类性能，而它们固定的掩码率则限制了在可变长度输入的生成任务中的泛化能力。尽管我们的任务不同于基于概率的分类，但结果表明，MLMs 可能通过双向上下文更有效地整合形式和含义。与这一趋势一致，一些 BabyLM 挑战系统（Hu 等，2024）已经采用了混合的 MLM-CLM 目标（Yu 等，2024；Charpentier 和 Samuel，2024）。对哪些语言能力特别受益于 MLM 目标的系统性探索仍然是未来工作的有前景方向。

5 Analysis

5 分析

5.1 Learning Trajectories

5.1 学习轨迹

Our work investigates the linguistic ability to interpret meanings implied by linguistic forms, an ability not captured by grammatical acceptability alone. This section traces the development of grammatical acceptability and constructional meaning understanding across training and data size. We hypothesize that grammatical acceptability improves earlier in training, as it often depends on relatively shallow statistical patterns. For instance, in number agreement, a model that learns to associate the number feature of the subject with the verb can correctly prefer “The cats annoy Tim.” over “*The cats annoys Tim.” In contrast, many constructions in CxMP require grasping the meaning encoded by the form itself, which likely demands more training data and emerges later in learning.

我们的工作探讨了语言解释语言形式所隐含意义的能力，这种能力不能仅通过语法可接受性来体现。本节追踪了语法可接受性和构式意义理解在训练和数据规模上的发展情况。我们假设语法可接受性在训练早期就会有所提高，因为这通常依赖于相对浅层的统计模式。例如，在数的一致性上，一个学会将主语的数特征与动词联系起来的模型，可以正确地偏好“这些猫让蒂姆烦恼。”而不是“* 这些猫让蒂姆烦恼。”相比之下，CxMP 中的许多构式需要理解形式本身所编码的意义，这可能需要更多的训练数据，并且在学习过程中较晚才会出现。

Settings We use the base versions of OLMo2-7B/13B, which provide publicly available checkpoints (OLMo et al., 2024). We select 11 representative checkpoints from Stage 1 of OLMo2-13B and several checkpoints from Stage 2. For OLMo2-7B, we adopt checkpoints that correspond to approximately the same cumulative token counts as those used for the 13B model. For eval-

uation Datasets, we use BLiMP (Warstadt et al., 2020) in addition to CxMP. BLiMP is a benchmark designed to evaluate the grammatical sensitivity of language models. It consists of pairs of (un)grammatical sentences in 12 linguistic phenomena, and we compute scores for each phenomenon. The scoring method is identical to that used for CxMP.

设置我们使用 OLMo2-7B/13B 的基础版本，这些版本提供了公开可用的检查点（OLMo 等，2024）。我们从 OLMo2-13B 的第一阶段中选择了 11 个具有代表性的检查点，并从第二阶段中选择了若干检查点。对于 OLMo2-7B，我们采用与 13B 模型使用的检查点大致对应的累计 token 数量。对于评估数据集，我们使用 BLiMP（Warstadt 等，2020）以及 CxMP。BLiMP 是一个用于评估语言模型语法敏感性的基准测试。它由 12 种语言现象中的（不）合法规则句子对组成，我们为每种现象计算得分。评分方法与 CxMP 相同。

Results Figure 6 shows the trajectories of BLiMP and CxMP scores (log scale) as a function of training token count. For BLiMP, the models correctly solve roughly 80% of linguistic phenomena by around 50B tokens, after which the scores plateau, showing only a minimal increase of about one to two points overall. In contrast, CxMP scores continue to improve gradually from around 10B tokens until the final stage of training. These results suggest that language models acquire formal linguistic knowledge, such as grammatical acceptability, relatively early in training, whereas mastering more complicated linguistic abilities, such as constructional meaning, requires longer-term learning. Traditionally, model performance has been primarily assessed with BLiMP, and reaching a certain score has often been taken as evidence of sufficient lin-

图 6 展示了 BLiMP 和 CxMP 分数（对数刻度）随训练标记数量变化的轨迹。对于 BLiMP，模型在约 500 亿个标记时能够正确解决大约 80% 的语言现象，之后分数趋于平稳，总体仅增加约 1 到 2 分。相比之下，CxMP 分数从约 100 亿个标记开始逐渐提高，直到训练的最后阶段。这些结果表明，语言模型在训练早期就获得了形式语言知识，如语法可接受性，而掌握更复杂的语言能力，如构式意义，则需要更长时间的学习。传统上，模型性能主要通过 BLiMP 进行评估，达到某一分数通常被视为语言能力充分发展的证据。

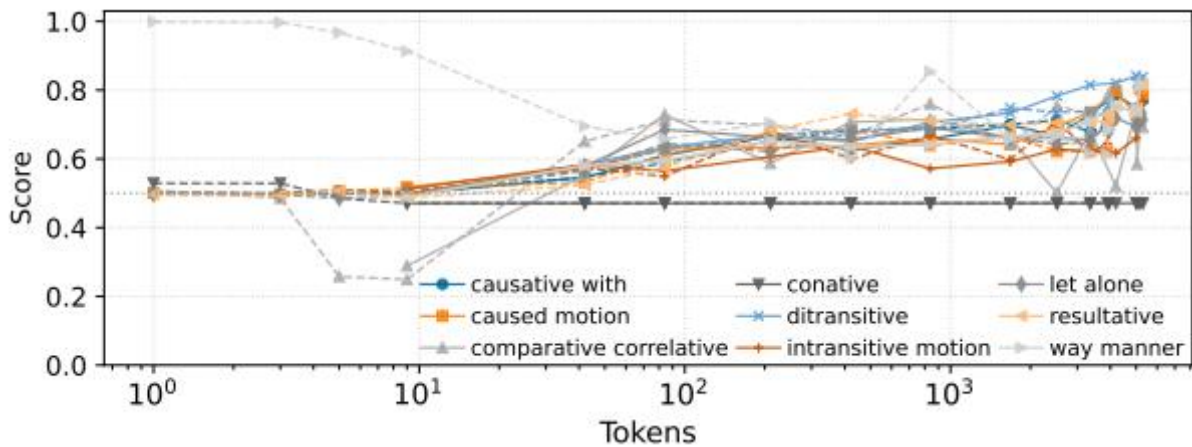


Figure 6: images/11ee20be751d069075de3f2280734650eb8d33e612593a330faae1041fd34937.jpg

(a) Scores on CxMP

(a) 在 CxMP 上的得分

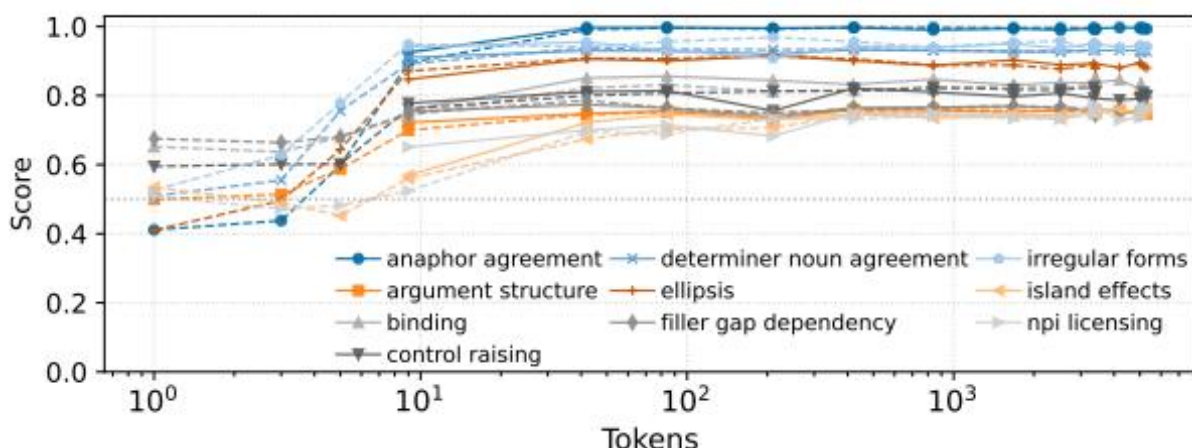


Figure 7: images/9d3867172dc05279dc2a21ee35b550d94ab92405dbb1cadde9fe0b343aafe0a3.jpg

(b) Scores on BLiMP

Figure 6: Scores of OLMo-2 on CxMP and BLiMP across training token counts (log scale). Solid/Dashed lines represent the 13B/7B model.

(b) 在 BLiMP 上的得分

图 6: OLMo-2 在 CxMP 和 BLiMP 上的得分随训练 token 数量（对数刻度）的变化。实线/虚线分别表示 13B/7B 模型。

guistic competence. However, the results from CxMP reveal that understanding not only the formal grammatical aspects of constructions but also their semantic and functional dimensions requires continued learning.

语言能力。然而，CxMP 的结果表明，不仅需要学习构式的形式语法方面，还需要持续学习其语义和功能维度。

5.2 Effect of Bias

5.2 偏见的影响

In Section 4.2, we find that models trained on developmentally appropriate data sizes and domains remain near chance level on many constructions and variants. In contrast, mid-sized models show large score gaps across variants, whereas larger models. This tendency is smaller in larger models. This pattern suggests the presence of biases driven by shallow heuristics. We hypothesize that such biases may appear not only between variants but also between switched and non-switched sentences. This section analyzes how these biases evolve throughout training, aiming to identify when such heuristic-driven behaviors begin to emerge.

在第 4.2 节中，我们发现，在适龄数据规模和领域上训练的模型在许多构造和变体上仍接近随机水平。相比之下，中等规模的模型在不同变体之间表现出较大的分数差异，而更大规模的模型则显示出这种趋势较小。这种模式表明存在由浅层启发式规则驱动的偏差。我们假设这种偏差不仅出现在不同变体之间，还可能出现在切换句与非切换句之间。本节分析这些偏差在整个训练过程中是如何演变的，旨在确定此类启发式驱动行为何时开始显现。

Definition of Bias (i) Bias between switched and non-switched sentences: This bias arises when scores differ between switched and non-switched sentences within the same construction and variant. For example, in “Gianna left Nancy with a receipt. The one who provided the receipt is Gianna/*Nancy.” the model correctly prefers the plausible evaluation pair. However, in

the switched version “Nancy left Gianna with a receipt. The one who provided the receipt is Nancy/*Gianna.” it still tends to choose “Gianna.” In such cases, the model

偏差的定义 (i) 切换句与非切换句之间的偏差: 当在同一结构和变体中, 切换句与非切换句的得分存在差异时, 就会出现这种偏差。例如, 在 “Gianna 把收据留给了 Nancy。提供收据的人是 Gianna/*Nancy。” 这个例子中, 模型正确地倾向于选择合理的评价对。然而, 在切换后的句子 “Nancy 把收据留给了 Gianna。提供收据的人是 Nancy/*Gianna。” 中, 模型仍然倾向于选择 “Gianna”。在这些情况下, 模型

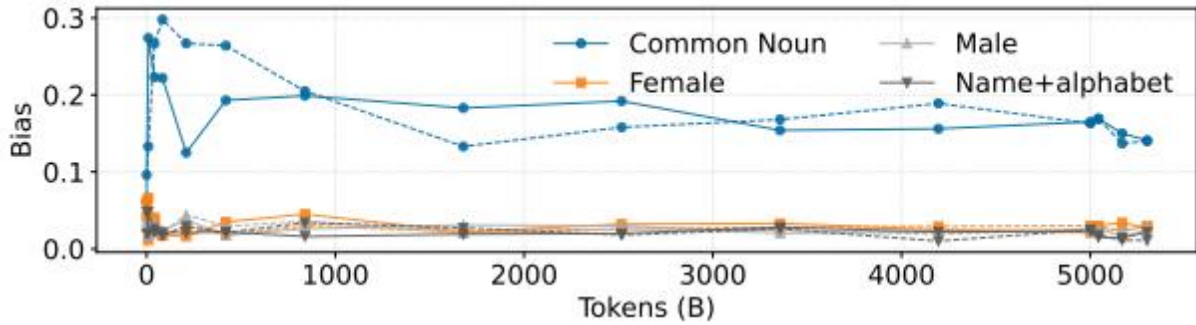


Figure 8: images/85e2684b4c24958eab24d52e323ee8d18555894ddcb36d3ccd0cbd0c842f744a.jpg

(a) Bias b/w switched and non-switched sentences (;i)

(a) 偏差 b/w 切换和非切换句子 (;i)

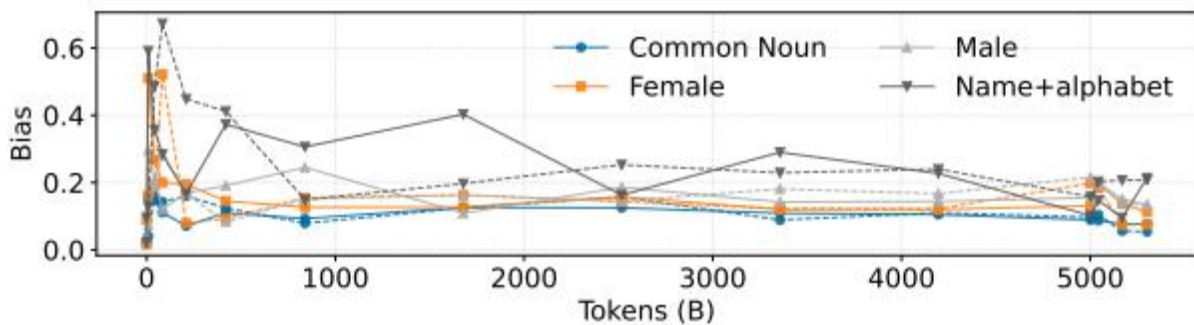


Figure 9: images/39206ae794c9b6a912b3f40196c4eeb5a1ddf5f6f117af6b0cb951e267d0a13b.jpg

(b) Bias across variants (;ii)

Figure 7: Biases of OLMo-2 on CxMP across training token counts (log scale).

(b) 不同变体的偏差 (;ii)

图 7: OLMo-2 在 CxMP 上随训练 token 数量 (对数刻度) 变化的偏差。

may rely solely on the diagnostic sentence, mechanically selecting the subject most frequently associated with the verb. This bias is likely more pronounced when the protagonists are common nouns, as switching reverses the typical agent-patient relation (e.g., The student taught the teacher English.). (ii) Bias across variants: This bias occurs when scores differ between variants of the same construction. For instance, while the model correctly prefers the plausible evaluation pair in “Gianna left Nancy with a receipt. The one who provided the receipt is Gianna/*Nancy.” it still favors “Gianna” in “Gianna left Nancy with a receipt. The one who received the receipt is Nancy/*Gianna.” That is, it selects the implausible evaluation pair. Such patterns suggest that the model may rely on shallow heuristics based on surface cues, such as choosing the nearest noun

or consistently selecting the first noun in the sentence.

可能仅依赖于诊断句子，机械地选择与动词关联最频繁的主语。这种偏差在主角是普通名词时可能更为明显，因为切换会颠倒典型的施事-受事关系（例如：The student taught the teacher English.）。(ii) 变体间的偏差：当同一结构的不同变体之间的得分不同时，就会出现这种偏差。例如，虽然模型在“Gianna left Nancy with a receipt. The one who provided the receipt is Gianna/*Nancy.”中正确地偏好合理的评价对，但在“Gianna left Nancy with a receipt. The one who received the receipt is Nancy/*Gianna.”中仍然倾向于选择“Gianna”。也就是说，它选择了不合理的评价对。这种模式表明，模型可能依赖于表面线索的浅层启发式策略，例如选择最近的名词或始终选择句子中的第一个名词。

Quantification of Bias We evaluate each checkpoint of the same models used in Section 5.1 using the following metrics. (i) Bias between switched and original sentences: For each construction and noun type, we computed the absolute difference between the mean scores of the original and switched variants (A vs. A_switched, B vs. B_switched). (ii) Bias across variants: For each construction and noun type, we calculate the absolute difference between the mean scores of the two variants (A/A_switched, B/B_switched), where each variant’s score was averaged over its original and switched forms. We then average these values across constructions within each noun type. Note that larger values would indicate stronger bias in the model.

偏差量化我们使用以下指标评估第 5.1 节中使用的每个模型的检查点。(i) 切换句与原句之间的偏差：对于每个结构和名词类型，我们计算原句和切换句的平均得分的绝对差异（A vs. A_switched, B vs. B_switched）。(ii) 不同变体之间的偏差：对于每个结构和名词类型，我们计算两个变体的平均得分的绝对差异（A/A_switched, B/B_switched），其中每个变体的得分在其原始形式和切换形式上的平均值。然后我们在每个名词类型内对这些值进行平均。需要注意的是，较大的值表示模型存在更强的偏差。

Results Figures 7a and 7b show the score trajectories across checkpoints (training token count) for each noun type, corresponding to Type (i) and Type (ii), respectively. In Type (i), this trend is most prominent for Common nouns, while in Type (ii), it appears more broadly, especially for Name+alphabet. In both cases, the scores start low in the early training stages, rise thereafter, and gradually decline again toward the end. Compared with the results in Section 5.1 (Figure 6a), this pattern suggests that the model initially responds at random without bias, then develops shallow heuristics, and finally converges toward more accurate judgments. As expected, in Type (i), this tendency is particularly evident for Common nouns. In contrast, in Type (ii), a similar pattern emerges most clearly for Name+alphabet. This may indicate that the model becomes confused by low-frequency input formats and overly relies on nearby cues. Overall, these results demonstrate that different kinds of biases can arise depending on the noun type and sentence structure being tested. They also highlight the importance of incorporating diverse question formulations to comprehensively assess the model’s linguistic knowledge.

图 7a 和 7b 展示了每种名词类型（分别对应类型 (i) 和类型 (ii)）在检查点（训练 token 数量）上的得分轨迹。在类型 (i) 中，这种趋势在普通名词中最为明显，而在类型 (ii) 中，这种趋势则更为广泛，尤其是在 Name+alphabet 中。在两种情况下，得分在早期训练阶段较低，随后上升，最终在接近结束时逐渐下降。与第 5.1 节的结果（图 6a）相比，这种模式表明模型最初以随机方式响应，没有偏见，随后发展出浅层次的启发式方法，最后逐渐趋于更准确的判断。正如预期的那样，在类型 (i) 中，这种趋势在普通名词中尤为明显。相比之下，在类型 (ii) 中，Name+alphabet 最清晰地表现出类似的模式。这可能表明模型对低频输入格式感到困惑，并过度依赖于附近的线索。总体而言，这些结果表明，根据被测试的名词类型和句子结构，可能会产生不同的偏见。它们还强调了纳入多样化的提问方式以全面评估模型语言知识的重要性。

5.3 CxMP Using Pseudoword

5.3 CxMP 使用伪单词

In theoretical linguistics, the meanings of constructions are theoretically defined and systematized (Goldberg, 1995). In contrast, Kako (2006) experimentally examines how speakers perceive semantic properties directly from constructions themselves. He uses stimulus sentences such as “The rom gorped the blickit to the dax.”, in which verbs and nouns are replaced with meaningless pseudowords, thereby controlling for lexical semantics and evaluating how humans inter-

pret meaning solely from constructional form. The results reveal significant correspondences between each construction and the semantic attributes theoretically assumed in prior work. In this section, adopting a similar perspective, we examine how well language models can capture the correspondence between constructions and their meanings in sentences where the influence of lexical semantics is mitigated. Specifically, using CxMP, we evaluate whether models can reconstruct meaning based on constructional form.

在理论语言学中，构式的意义被理论性地定义和系统化 (Goldberg, 1995)。相比之下，Kako (2006) 通过实验研究了说话者如何直接从构式本身感知语义属性。他使用诸如 “The rom gorpmed the blickit to the dax.” 这样的刺激句，其中动词和名词被替换为无意义的伪词，从而控制词义语义的影响，并评估人类如何仅根据构式形式来理解意义。结果揭示了每个构式与先前研究中理论假设的语义属性之间存在显著的对应关系。在本节中，我们采用类似的视角，考察语言模型在削弱词义语义影响的句子中，能否很好地捕捉构式与其意义之间的对应关系。具体而言，我们使用 CxMP，评估模型是否能够基于构式形式重建意义。

Settings We create the evaluation dataset used in the experiment by replacing the content words in CxMP with pseudowords. Our analysis focuses on

设置我们通过将 CxMP 中的实词替换为伪词来创建实验中使用的评估数据集。我们的分析重点在于

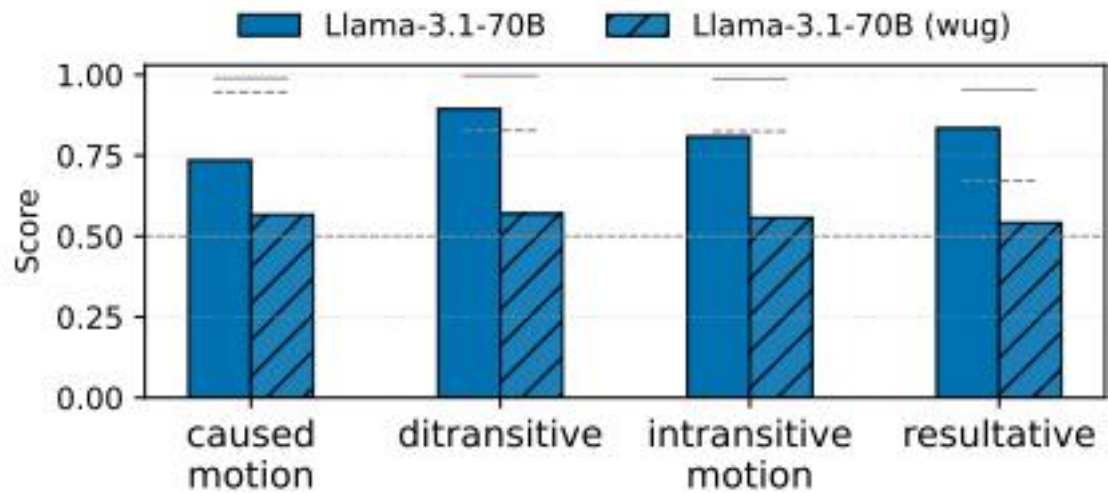


Figure 10: images/262a2d87fedcf70cb2ade76ad8f467422c0a0c546da15a19286f37c9683b906a.jpg

Figure 8: Scores for pseudowords (wug) and real words.

图 8：假词（wug）和真实单词的评分。

four major argument-structure constructions identified by Goldberg (Goldberg, 1995): CAUSED-MOTION, INTRANSITIVE-MOTION, DITRANSITIVE, and RESULTATIVE. Among these, DITRANSITIVE and INTRANSITIVE-MOTION overlap with the constructions examined by Kako (2006). We generate the pseudowords with Wuggy (Keuleers and Brysbaert, 2010), a tool that produces phonotactically natural words (wugs). The detailed settings are in Appendix H. We use Llama3.1-70B as the target model.

由 Goldberg (Goldberg, 1995) 识别出的四种主要论点结构构式：CAUSED-MOTION、INTRANSITIVE-MOTION、DITRANSITIVE 和 RESULTATIVE。其中，DITRANSITIVE 和 INTRANSITIVE-MOTION 与 Kako (2006) 所研究的构式有重叠。我们使用 Wuggy (Keuleers 和 Brysbaert, 2010) 生成伪词，这是一种能够生成语音上自然的词（wugs）的工具。详细设置见附录 H。我们使用 Llama3.1-70B 作为目标模型。

Results Figure 8 shows the scores for each construction using either pseudowords (wugs) or the original words. Across all constructions, scores drop markedly when original words are replaced with pseudowords, indicating that the model struggles to infer meaning solely from construc-

tional cues without relying on lexical content. Nevertheless, scores remain slightly above chance, suggesting that the model retains some sensitivity to constructional information. By construction type, Kako (2006) reported that the presence or absence of closed-class words (e.g., to, that) had little effect on performance. Among the constructions tested here, CAUSED-MOTION and INTRANSITIVE-MOTION include such closed-class words, and their pseudoword–original differences are smaller than those of other constructions, with overall scores remaining stable. The pseudowords used are phonotactically natural and morphologically plausible, potentially sharing letters with the original words and thus offering weak cues. Following prior work, we also retained determiners and inflectional suffixes such as -ed and -ing, which may provide part-of-speech cues that facilitate construction identification. Future work should further examine which linguistic elements contribute to identifying constructions under varying levels of lexical control.

图 8 展示了每种构式使用假词 (wugs) 或原词时的得分情况。在所有构式中, 当原词被假词取代时, 得分显著下降, 这表明模型在不依赖词汇内容的情况下, 难以仅通过构式线索推断含义。然而, 得分仍略高于随机水平, 表明模型在一定程度上仍对构式信息保持敏感。根据 Kako (2006) 的报告, 闭类词 (如 to、that) 的存在与否对表现影响不大。在本研究测试的构式中, CAUSED-MOTION 和 INTRANSITIVE-MOTION 包含此类闭类词, 其假词与原词之间的差异小于其他构式, 总体得分保持稳定。所使用的假词在语音上自然且形态上合理, 可能与原词共享字母, 从而提供较弱的线索。根据先前的研究, 我们还保留了如-ed 和-ing 等限定词和屈折后缀, 这些可能提供词性线索, 有助于构式的识别。未来的工作应进一步探讨哪些语言成分在不同词汇控制水平下有助于识别构式。

6 Conclusion

6 结论

We present CxMP, a Construction Grammar-based benchmark for evaluating how language models

我们提出 CxMP, 这是一个基于构造语法的基准, 用于评估语言模型如何处理语言。

grasp meanings encoded by constructions. Results indicate that there remains considerable room for improvement in constructional meaning understanding, not only in developmentally plausible small models but also in modern LLMs.

把握由结构编码的意义。结果表明, 在结构意义理解方面仍有很大的改进空间, 不仅在发展上合理的较小模型中如此, 现代的大型语言模型 (LLMs) 也是如此。

Limitations

限制

While CxMP provides a controlled and interpretable framework for evaluating constructional understanding in language models, several limitations remain. First, although the dataset covers a broad range of constructions, it focuses primarily on English and on a limited subset of construction types. Extending the benchmark to other languages or to more diverse construction families would allow testing whether the observed developmental patterns generalize across typologically different systems. Second, the diagnostic design relies on written, isolated sentences. Real-world language understanding involves richer discourse and pragmatic contexts, which may influence how constructions are interpreted. Incorporating contextualized or multi-turn settings would improve ecological validity. Third, while pseudoword experiments mitigate lexical bias, they may also introduce unnatural stimuli that underestimate model competence. Future work could explore alternative methods for lexical control that preserve naturalness. Finally, although CxMP evaluates a wide range of models, it remains static. Coupling this benchmark with dynamic evaluation protocols, such as probing internal representations or testing fine-tuned adaptations,

could reveal deeper insights into how constructional meaning emerges and is represented within models.

尽管 CxMP 为评估语言模型中的构式理解提供了一个可控且可解释的框架，但仍存在一些限制。首先，虽然该数据集涵盖了广泛的构式类型，但主要集中在英语以及构式类型的一个有限子集上。将基准扩展到其他语言或更多样化的构式家族，将有助于测试所观察到的发展模式是否能在类型学上不同的系统中推广。其次，诊断性设计依赖于书面的、孤立的句子。现实中的语言理解涉及更丰富的语篇和语用背景，这些背景可能会影响构式的解释方式。纳入语境化或多轮对话设置将有助于提高生态效度。第三，尽管伪词实验可以缓解词汇偏差，但它们也可能引入不自然的刺激，低估模型的能力。未来的工作可以探索替代的词汇控制方法，以保持自然性。最后，尽管 CxMP 评估了广泛范围的模型，但它仍然是静态的。将此基准与动态评估协议相结合，例如探测内部表示或测试微调适应，可以揭示构式意义如何在模型中产生和表示的更深层次见解。

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